Nonlinear Practice 3

Module 8 - Nonlinear Pharmacokinetics · Daily practice sheet · No date

Name _____ Date _____

1. A 78-kg, 28 year old male is receiving phenytoin for the treatment of seizures. When this patient was taking a daily oral dose of 300 mg, his steady state plasma concentration was 10.4 mg/L. The dose was increased to 400 mg which resulted in a steady-state plasma concentration of 22 mg/L. Assume 100% bioavailability and an apparent volume of distribution that is equal to 70% of body weight.

a Calculate KM

b Calculate Vmax

c Calculate the half-life and clearance in this patient at steady-state while he is taking 300 mg/day. What is the half-life?

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d Calculate the half-life and clearance in this patient at steady-state while he is taking 400 mg/day. What is the half-life?

d Calculate the half-life and clearance in this patient at steady-state while he is taking 400 mg/day. What is the clearance?

e What is the daily phenytoin dose that should achieve a steady-state phenytoin plasma concentration of 15 mg/L in this patient? Please round to the nearest 10 mg.